From Truth Table to Quantum Circuit: K-Map-Guided Toffoli Synthesis

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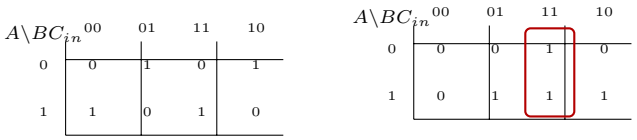
***Abstract*—Karnaugh-map grouping is unsound when applied unchanged to reversible circuits: a classical AND-OR circuit *sets* its output, but a Toffoli circuit *flips* it, so a cell covered by two groups is flipped back to 0. We prove that a Toffoli circuit built from cubes is correct exactly when every cell is covered an odd number of times if the function is 1 there and an even number of times otherwise. Disjoint covers, grown greedily on the K-map, are the simplest such covers; fixed-polarity Reed–Muller forms and a term-merging rule give overlapping ones. Three function-preserving passes, each with a proof, shrink every candidate: deriving one output from another, cancelling adjacent X gates, and reordering terms so that more X gates cancel. Every candidate is checked by exhaustive simulation, transpiled to a model of a 133-qubit IBM device and checked again, and the one with the fewest CZ gates is kept. On eleven functions of two to eight inputs, disjoint covers alone are not competitive: tweedledum's ESOP circuits have fewer gates on eight functions and need fewer CZ gates on seven. With overlapping covers, our circuits need fewer CZ gates than tweedledum's better circuit on six functions, as many on four, and more on one (`3_17`, 49 vs. 44). Published in-place circuits for `3_17` and `rd32` remain $2.0\times$ and $1.5\times$ smaller.**

***Index Terms*—reversible logic synthesis, Karnaugh maps, Toffoli gates, disjoint cube cover, ESOP, Reed–Muller, quantum circuit synthesis, quantum cost**

I. Introduction

Grover oracles must flip a target qubit iff a classical predicate $f(x) = 1$, and Shor-style arithmetic chains one-bit adders; both need correct reversible circuits built from truth tables. The textbook tool for minimizing such logic, the Karnaugh map [1], allows overlapping groups. That is sound for a gate that sets its output but not for one that XORs into it: for $\mathrm{Out} = A + B$, the naive circuit `cx(B,Out); cx(A,Out)` returns 0 on $A{=}B{=}1$.

Our contributions are: (1) a parity condition under which any cube cover gives a correct Toffoli circuit, with a greedy algorithm for exact disjoint covers on the K-map and overlapping covers from fixed-polarity Reed–Muller forms and term merging; (2) output derivation, X-cancellation, and term reordering, each proved function-preserving, with derivation chosen by the gate count left after cancellation; (3) selection among checked candidates by CZ count on an IBM device model, and an evaluation on eleven functions against published RevLib circuits and tweedledum, which reports where our circuits are larger. The code that produces every number here is at https://github.com/ArhamAkhtarAlam/toffoli-synthesis-final-paper.



Sum: no two 1-cells adjacent (parity).

Carry: group BC_{in}; the other two 1-cells must be single terms.

Fig. 1. K-maps for the full adder. No two of Sum's 1-cells are adjacent; Carry's disjoint cover is the boxed pair and two single cells.

II. Related Work

Reversible synthesis from truth tables [2] includes transformation-based synthesis [3], ESOP-based synthesis (e.g. EXORCISM-4 [4]), BDD-based synthesis [5], and exact SAT-based synthesis [6]. Our covers are ESOPs: the disjoint ones are a restricted ESOP with a one-line correctness proof, and fixed-polarity Reed–Muller forms [2] are a classical source of overlapping ones (tweedledum's PPRM is the positive-polarity case). RevLib [7] collects benchmarks and published circuits; RevKit [8] and tweedledum [9] implement these synthesis families, and Qiskit [10] exposes truth-table oracles; its transpiler maps our circuits to a device's gates and connectivity.

III. Covers and Circuits

A *pattern* $p \in \{0, 1, -\}^n$ denotes the cube $[\![p]\!] = \{x : x_k = p_k \text{ whenever } p_k \neq -\}$. Circuits act on n input, m output, and a ancilla wires (outputs and ancillas start at 0) with gates X and C^kX (flip the target iff all k controls are 1): the positive-control MCT library of RevLib [7]. A term p becomes one C^kX on the wires where p is specified, wrapped in X gates on those where $p_k = 0$, so it flips its target iff $x \in [\![p]\!]$. The quantum cost of a C^kX is $2^{k+1} - 3$ (1 for X and CNOT) [11], [7].

A K-map lays out a truth table so that adjacent cells differ in one bit; a group of 2^d adjacent 1-cells is a cube with d dashes. Classical grouping lets groups overlap, but a Toffoli circuit flips a cell once per group containing it, so a cell in two groups ends at 0 (cell 11 of $A + B$). Overlaps are harmless only when every cell is covered an odd number of times, as for Carry's three pairs, which meet only at 111 (Theorem 2). Algorithm 1 avoids overlaps: it grows each group only inside the cells not yet covered (Fig. 1).

Algorithm 1 DISJOINTCOVER(S)

$R \leftarrow S$; $T \leftarrow \langle\rangle$
while $R \neq \emptyset$ **do**
  $p \leftarrow \min(R)$
  for $i \leftarrow 1$ **to** n **do** ▷ GROWCUBE
    if $[\![p$ with $p_i{=}-]\!] \subseteq R$ **then** $p_i \leftarrow -$
  end for
  append p to T; $R \leftarrow R \setminus [\![p]\!]$
end while
return T

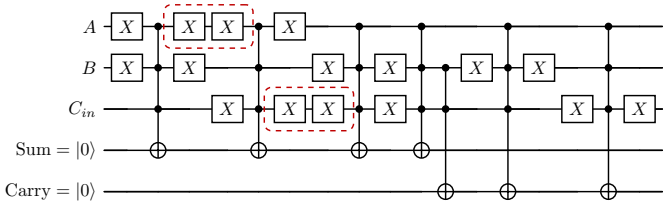


Fig. 2. Full adder as emitted: Sum's four terms, then Carry's three (23 gates). In each dashed box, one term ends and the next begins with an X on the same wire, and no gate between them uses that wire as a control; since $XX = I$, both are deleted (Lemma 3), leaving 19 gates.

Theorem 1 (Exact, disjoint cover)**.** *The cubes returned by Algorithm 1 are pairwise disjoint and their union is S.*

Proof. Each cube is grown only while it stays inside the current remainder R, which excludes all earlier cubes, so cubes are disjoint and lie in S. Each iteration removes at least its seed, so the loop ends, with $R = \emptyset$, after at most $|S|$ iterations. □

Theorem 2 (Correct synthesis)**.** *Emitting terms $p_1, \ldots, p_t$ onto output wire j leaves $f_j(x)$ on it, for every input x, iff $|\{i : x \in [\![p_i]\!]\}|$ is odd exactly when $f_j(x) = 1$. In particular,* DISJOINTCOVER(f_j) *does.*

Proof. A term flips wire j iff x is in its cube, and flips add mod 2. No gate targets an input wire. Disjoint cubes covering the on-set (Theorem 1) contain each on-cell once and each off-cell never. □

Fig. 2 shows the full adder: 4 singleton terms for Sum (parity admits no larger cube) and 3 for Carry, 23 gates in all.

Overlapping covers. For a polarity $\pi \in \{0,1\}^n$, the fixed-polarity Reed–Muller (FPRM) form writes f_j as an XOR of products of the literals x_i ($\pi_i = 0$) or $\bar{x}_i$ ($\pi_i = 1$); its coefficients follow from the truth table by a butterfly transform in $O(n2^n)$. Each product is a cube, and the form is an identity, so each of the 2^n FPRM forms satisfies Theorem 2 ($\pi = 0^n$ is the PPRM). For the full adder, $\mathrm{Sum} = A \oplus B \oplus C_{in}$ is three CNOTs instead of four 3-control terms.

Lemma 1 (Merging)**.** *Two cubes that differ in one position i can be replaced by the cube with the third value of $\{0, 1, -\}$ at i; two equal cubes can be deleted. Neither changes the parity in Theorem 2.*

Proof. With c the common part, $cx_i \oplus c\bar{x}_i = c$, $c \oplus cx_i = c\bar{x}_i$, $c \oplus c\bar{x}_i = cx_i$, and $c \oplus c = 0$. □

Merging removes a term but can add negated literals (X gates), so both versions of each cover are kept. The candidate covers of an output are Algorithm 1 under every variable order ($2n$ rotations and reversals for $n > 4$) with the smallest or largest seed, all FPRM forms, and each of these merged.

IV. OPTIMIZATIONS

Derivation. A later output f_k may be built as a CNOT from an earlier output wire j followed by a cover of $r = f_j \oplus f_k$.

Lemma 2 (Derivation)**.** *If wire j holds $f_j(x)$, a CNOT $j \to k$ followed by the terms of a cover of r that satisfies Theorem 2 leaves $f_k(x)$ on k and changes no other wire.*

Proof. Wire k ends at $f_j(x) \oplus r(x) = f_k(x)$ by Theorem 2 applied to r. □

On a two-bit ripple adder ($S_0, C_0, S_1, \mathrm{Cout}$) derivation finds $S_1 \oplus C_0 = A_1 \oplus B_1$, the identity "sum = inputs XOR carry," and cuts its disjoint circuit from 126 gates to 64. A term common to several outputs can also be computed once into an ancilla, copied, and uncomputed; for two outputs this costs $2c + 2$ gates against $2c$, so it almost never pays and no circuit here uses it.

X-cancellation. Adjacent terms often leave $X_w \cdots X_w$ on an input wire w, as in the two boxed pairs of Fig. 2.

Lemma 3 (X-cancellation)**.** *If no gate between two X_w uses w as a control, deleting both leaves the circuit's action unchanged.*

Proof. X_w commutes with gates not involving w and with gates targeting w (both flip w; their condition does not depend on w), so the two X_w become adjacent and $X_w X_w = I$. □

Since X gates sit only on inputs, the cancelled length is the number of C^kX gates plus, per wire, the parity of the X gates between consecutive control uses, computed in one pass. Chosen by gate count *before* cancellation, derivation can hurt: the full adder goes from 19 (cancellation alone) to 22. We therefore score each derivation candidate by the cancelled length of the circuit so far, and fall back to the plain cover whenever it is shorter, so the pipeline is never worse than cover plus cancellation. On the ripple adder this derives Cout instead of S_0 from C_0: one gate more before cancellation, but 47 gates after it instead of 58.

Term ordering. Call each term's gates, and each derivation CNOT, a *block*.

Lemma 4 (Reordering)**.** *Any block order in which each derivation CNOT from output i follows every block writing wire i computes the same function.*

Proof. Each block XORs a function of wires it does not write into the wires it writes, so two blocks commute unless one writes a wire the other reads; only a derivation CNOT reads a non-input wire. Any order meeting the constraint is reached

TABLE I

GATES AFTER X-CANCELLATION (RAW: ONE TERM PER ON-SET ROW, BEFORE CANCELLATION; DISJOINT COVER; DISJOINT PIPELINE WITH TERM ORDERING; HYBRID), AND GATES, QUANTUM COST, AND CZ COUNT ON THE IBM HERON MODEL FOR THE DISJOINT PIPELINE (DISJ.), THE HYBRID (HYB.: THE CHECKED CANDIDATE WITH THE FEWEST CZ GATES), AND TWEEDLEDUM'S PKRM AND PPRM CIRCUITS, ALL ON THE SAME $n+m$ LINES.

			This work: gates				tweedledum gates		Q-cost				CZ on device			
Function	n	m	Raw	Cover	Disj.	Hyb.	PKRM	PPRM	Disj.	Hyb.	PKRM	PPRM	Disj.	Hyb.	PKRM	PPRM
AND	2	1	1	1	1	1	1	1	5	5	5	5	7	7	7	7
OR	2	1	7	4	4	4	4	3	8	8	8	7	7	7	10	8
XOR	2	1	6	6	6	2	2	2	14	2	2	2	17	2	2	2
NAND	2	1	11	6	6	2	6	2	10	6	10	6	7	7	8	7
Half adder	2	2	7	7	6	3	3	3	14	7	7	7	19	12	14	14
Full adder (`rd32`)	3	2	26	19	14	6	10	6	74	18	22	18	133	34	34	36
2-bit ripple adder	5	4	332	86	31	16	34	18	411	104	142	118	514	206	243	231
BCD to Excess-3	4	4	122	36	22	24	45	32	190	140	257	292	279	197	388	493
3×3 multiplier	6	6	615	123	60	27	80	27	1920	627	936	919	2058	724	1235	1039
4-bit equality	8	1	144	76	52	24	24	81	8180	472	472	1009	7534	699	752	1768
`3_17`	3	3	48	22	13	12	15	14	37	28	31	30	65	49	44	49

from the emitted one by swapping adjacent commuting blocks. □

Our heuristic builds an order greedily (append the admissible block that gives the shortest cancelled prefix), then moves single blocks while the cancelled length drops. For reference, dynamic programming over (placed blocks, per-wire X parity) gives the exact optimum in $O(2^B 2^n B)$ for B blocks; for the full adder's cover it matches exhaustive search over all 7! orders (15 gates).

Selection. Each output, and each derivation difference r, takes its cheapest candidate cover, by gates or by quantum cost after cancellation, from three pools: disjoint covers; all covers; disjoint and FPRM covers in merged form only. Each of these six rules, and the disjoint pipeline itself, yields a circuit after derivation, cancellation, and ordering; each is checked on all inputs and transpiled, and the one with the fewest CZ gates is kept. The disjoint pipeline is a candidate, so the hybrid is never worse on the device.

V. EVALUATION

Setup. The method is implemented in Python. The functions are the six built-in presets (AND, OR, XOR, NAND, half and full adder), a two-bit ripple adder with all four internal signals as outputs, BCD-to-Excess-3, a 3×3-bit multiplier, a 4-bit equality comparator (the shape of a Grover oracle predicate), and RevLib's `3_17` [7]. Every circuit, before and after each pass, is checked against its truth table on all 2^n inputs by classical replay, which is exact because every gate maps basis states to basis states. Algorithm 1 was also checked on 2000 random on-sets ($n \leq 6$) with no overlap or gap. Over 2000 random truth tables ($2 \leq n \leq 4$, $1 \leq m \leq 3$), cancellation shortens 47.2% of cover circuits (9.7% on average); without the post-cancellation choice the pipeline is worse than cover plus cancellation in 2.4% of them, and with it in none; ordering removes a further 9.4% and reaches the exact optimum in 94.9% of the 1904 circuits where that optimum was computed. Derivation is used in 48.7% of the 1326 tables with two or more outputs, cutting gates before cancellation by 22.5% when used; sharing is chosen in 0.4%. The checker accepted every circuit and rejected each one against its table with one output bit flipped. Choosing, without a transpiler, the candidate with the fewest gates among those no worse than the disjoint pipeline in gates and quantum cost, the hybrid is smaller than the disjoint pipeline on 76.3% of the 1907 non-empty tables (22.6% fewer gates and 25.1% lower quantum cost on average) and larger on none; other disjoint covers alone help on 11.1% (1.1%).

Disjoint pipeline. Table I shows the passes compounding: the multiplier goes from 615 to 60 gates. On the comparator no cube larger than a minterm exists, so the whole gain ($144 \to 52$) comes from cancellation and ordering; ordering stops at 52 there against an exact 48.

tweedledum. We ran tweedledum 1.1.1 [9]'s PKRM and PPRM synthesizers, which build each output as an ESOP on its own line, rewrote negative controls as X-sandwiches, applied our cancellation pass, and verified each circuit. Its better circuit has fewer gates than our disjoint pipeline on eight functions; ours has fewer only on BCD-to-Excess-3 (22 vs. 32) and `3_17` (13 vs. 14), and lower quantum cost only on BCD-to-Excess-3. The comparator is the worst case (Q-cost 8180 vs. 472): each of its disjoint terms is an 8-control minterm, while an ESOP shares structure across terms.

Hybrid. Allowing overlaps closes most of the gap. Choosing each output's cover by gate count, without derivation, the ripple adder, multiplier, and `3_17` come out at exactly PPRM's 18, 27, and 14 gates; derivation takes them to 16, 25, and 12. Table I lists the hybrid circuit with the fewest CZ gates: it has fewer gates than tweedledum's better circuit on three functions, as many on seven, and one more on OR. Fewer gates do not always mean fewer CZ gates: the 3-gate OR needs 9 against the 4-gate circuit's 7, and the 25-gate multiplier 1012 against the 27-gate one's 724, so selection keeps the latter.

Published circuits. For the bijection `3_17`, RevLib lists a Miller–Maslov–Dueck circuit [3] with 6 gates (Q-cost 14) on 3 lines and an exact elementary-gate circuit with Q-cost 10 [6]; tweedledum's transformation-based synthesis gives 7 gates (Q-cost 15). For `rd32`, the full adder, the best listed

MCT circuit has 4 gates (Q-cost 12) on 4 lines. Our disjoint circuits are $2.2\times$ and $3.5\times$ larger in gates and $2.6\times$ and $6.2\times$ in quantum cost, the hybrid ones $2.0\times$ and $1.5\times$ in both, partly because they are out-of-place ($n+m$ lines, inputs preserved) while the published ones are in-place or keep garbage. These circuits are quoted from RevLib, not re-run.

On a device. We transpiled every circuit with Qiskit 2.6 to FakeTorino, Qiskit's offline model of the 133-qubit IBM Heron processor `ibm_torino` (native CZ, $\sqrt{X}$, R_z; its coupling map), at optimization level 3 with a fixed seed, and checked each transpiled circuit, placement and routing included, by statevector simulation on all inputs. The transpiler must be told that qubits do not start in $|0\rangle$: by default it borrows untouched input wires as clean scratch qubits, and the transpiled ripple adder is then wrong on some inputs. Our disjoint circuits need fewer CZ gates than tweedledum's better one only on OR and BCD-to-Excess-3 (279 vs. 388), tie on AND and NAND, and need more on the other seven. Which circuit needs fewer CZ gates matches which has lower quantum cost in 18 of the 22 comparisons, and which has fewer MCT gates in only 13: on `3_17` ours has fewer gates but 65 CZ gates against 44. Quantum cost is the better proxy. The hybrid needs fewer CZ gates than tweedledum's better circuit on six functions (e.g. 206 vs. 231 for the ripple adder, 724 vs. 1039 for the multiplier, 197 vs. 388 for BCD-to-Excess-3), as many on AND, XOR, NAND, and the full adder, and more only on `3_17` (49 vs. 44).

VI. Limitations and Conclusion

A Toffoli circuit built from cubes is correct exactly when every cell's coverage has the right parity (Theorem 2). This makes Karnaugh grouping sound with disjoint groups, with a proof short enough to check by hand, and admits overlapping covers from Reed–Muller forms and merging; derivation, cancellation, and reordering keep the guarantee (Lemmas 1–4). Disjoint covers alone are not competitive with tweedledum on gate count, quantum cost, or device CZ count. With overlapping covers and selection by CZ count, our circuits need no more CZ gates than tweedledum's better circuit on ten of eleven functions, but published in-place circuits are still smaller. The choices remain greedy: each output's cover is chosen on its own, derivation bases one at a time, and the ordering misses the optimum in 5.1% of random trials. Next are an in-place variant, ESOP search beyond FPRM forms [4], and selection without a transpiler in the loop. We did not run RevKit, the device results use one transpiler, one device model, and one seed, and all functions have $n \leq 8$.

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